



GOSFORD HIGH SCHOOL

NAME:

Task 2 June 2012
YEAR 11 Preliminary MATHEMATICS

Multiple Choice	/ 7
Part A Geometry	/16
Part B Linear Functions	/16
Part C Quadratic Polynomials	/16
Total	/55

General Instructions:

- Reading time – 5minutes
- Working time – 80 minutes
- Write using black or blue pen.
- Board-approved calculators may be used
- All necessary working should be shown in every question.

Total marks: - 120

- Attempt ALL Questions
- Answer Multiple choice questions on paper provided.
- Part A, B and C on your paper (start each part on a new sheet).

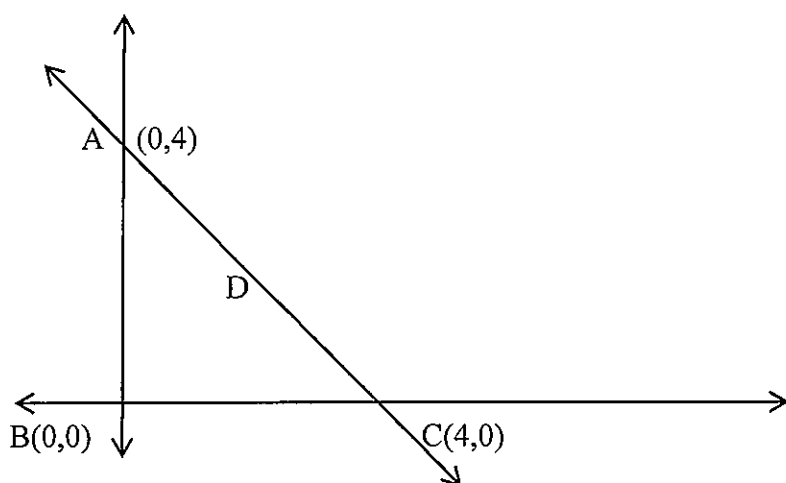
Multiple Choice Questions

Answers to the multiple choice on the answer sheet provided. (1 Mark Each)

1) A quadrilateral has equal diagonals that bisect each other at right angles, it is a

- (A) Rectangle
- (B) Rhombus
- (C) Isosceles Trapezium
- (D) Square

2) The following figure shows an isosceles triangle ABC. If D is the mid-point of AC, what is the length of AD?

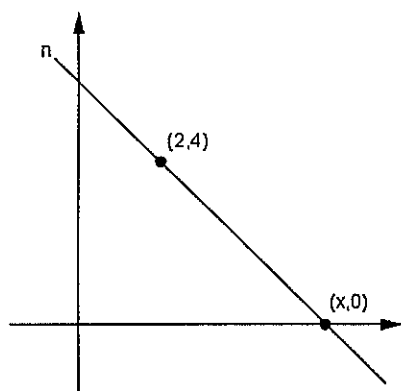


- (A) $2\sqrt{2}$
- (B) $\sqrt{2}$
- (C) $2 + \sqrt{2}$
- (D) $2 - \sqrt{2}$

3) Two functions are defined as: $f(x) = 3x^2 - 4$ and $g(y) = y^2 - 2y$. How many values of the variable a satisfy the equation $(a) = g(2a)$?

- (A) One
- (B) Two
- (C) Three
- (D) None

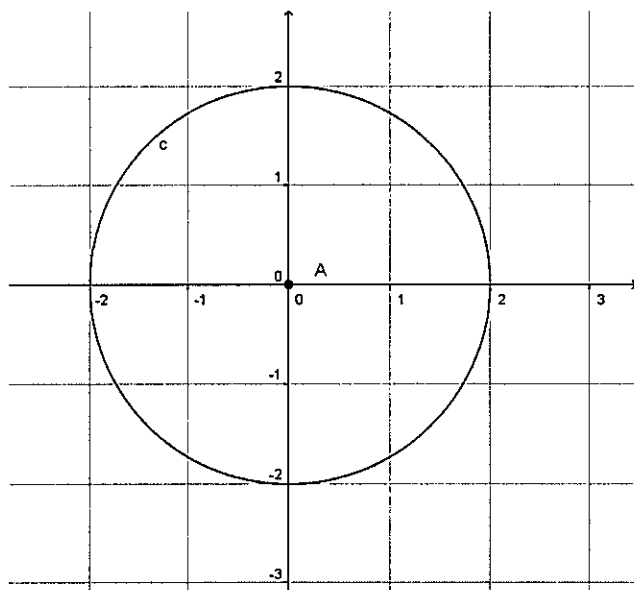
4)



In the above figure, the slope of line n is -1 . Find the value of x

- (A) 3
- (B) 4
- (C) 5
- (D) 6

5)



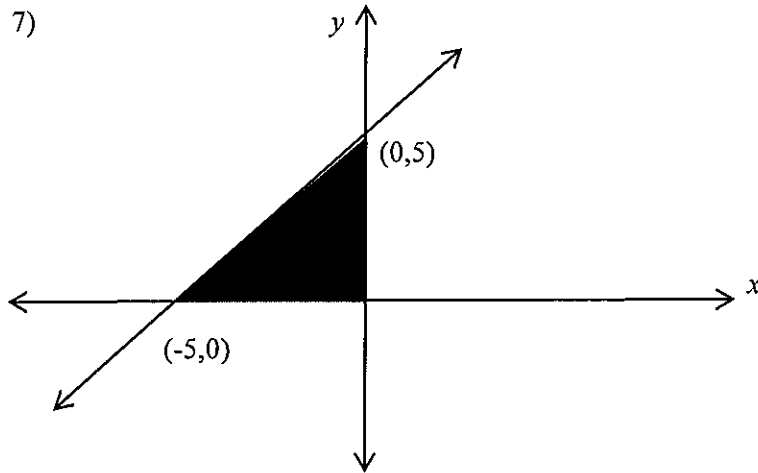
The above figure shows a circle drawn in an X-Y plane with centre A as the origin. If a line with equation $x + y = k$ intersects the circle at two points, which of the following is NOT a possible value of k ?

- (A) 0
- (B) $-2\sqrt{2} + 1$
- (C) 2
- (D) $2\sqrt{2} + 1$

6) Which of the following quadratic equations does **not** have real roots?

- (A) $y = 3x^2 - 3x - 2$
- (B) $y = 3x^2 + 3x - 2$
- (C) $y = 3x^2 - 3x + 2$
- (D) $y = 3x^2 + 3x - 2$

7)



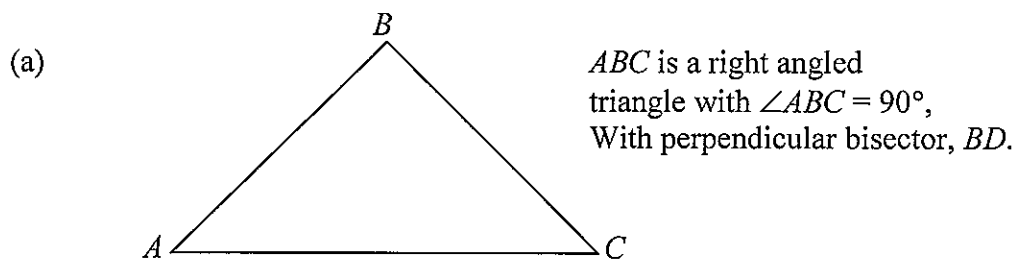
In the above diagram, the shaded region is given by:

- (A) $x \leq 0, y \leq 0$ and $x - y \leq -5$
- (B) $x \geq 0, y \geq 0$ and $y \leq x - 5$
- (C) $x \leq 0, y \geq 0$ and $x + y - 5 \leq 0$
- (D) $x \leq 0, y \geq 0$ and $y \leq x + 5$

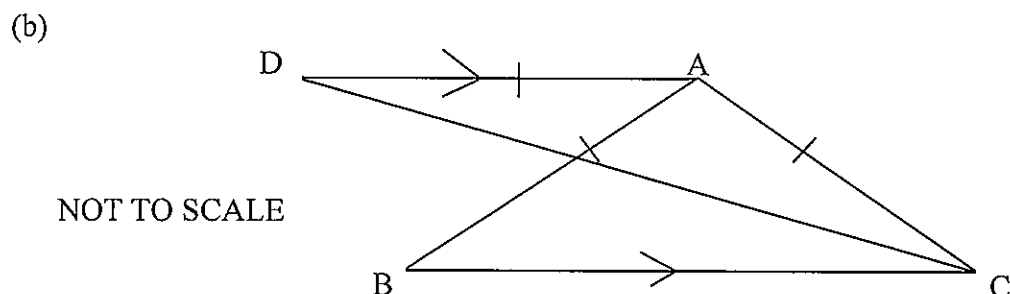
End of Multiple Choice

PART A Start your answers on a new page.

16 Marks



- (i) Copy diagram onto your paper indicating perpendicular bisector and showing all given information. (1)
- (ii) Prove triangles ABC and BDC are similar (3)
- (iii) Hence, show that $BC^2 = AC \times DC$ (1)

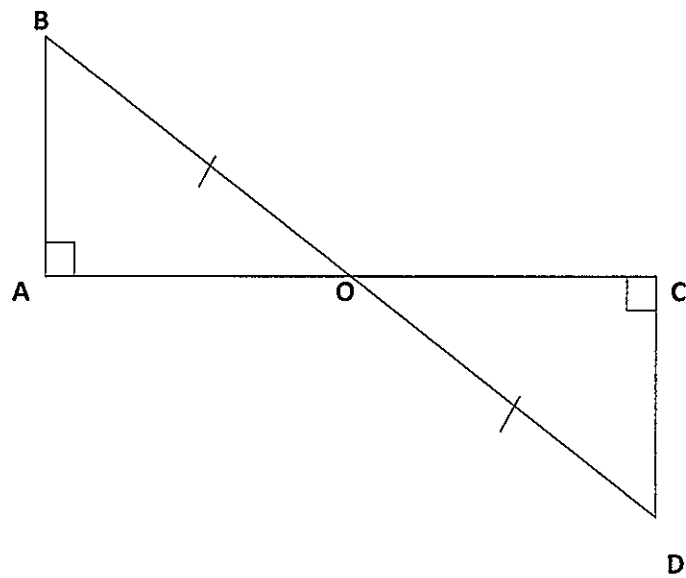


In the diagram, $\triangle ABC$ is isosceles with $AB=AC$.
 DA is parallel to BC and $DA=AC$ with $\angle DAB = 55^\circ$.

- (i) Copy the diagram onto your worksheet clearly
Showing all of the information. (1)
- (ii) Show that $\angle ACB = 55^\circ$ (2)
- (iii) Find size of $\angle ADC$ (2)
- (iv) Find the size of $\angle BCD$ (1)

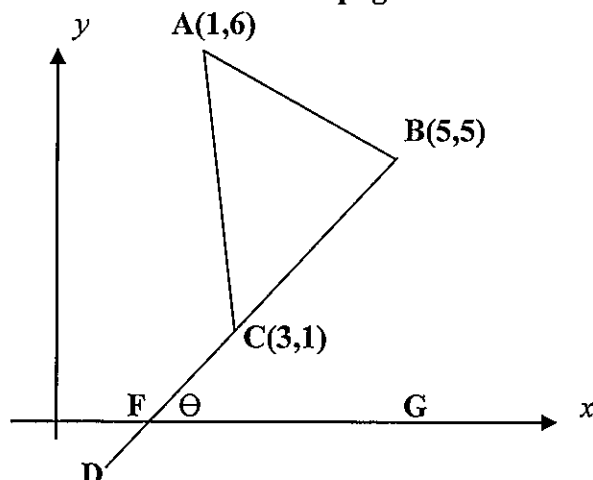
PART A Contd.

(c)



Prove that $\triangle AOB$ is congruent to $\triangle COD$. (3)

(d) Find the size of each exterior angle of a regular hexagon (2)

PART B**Start a new page.****16 Marks****NOT TO SCALE**

The points A , B and C have co-ordinates $(1,6)$, $(5,5)$ and $(3,1)$. The points B , C , F and D are in a straight line, and angle BFG is θ .

- (a) Copy the diagram and answer the following.
- (b) Find the gradient of the line BC (1)
- (c) Calculate the size of angle θ to the nearest degree (1)
- (d) Find the equation of line BD in general form (2)
- (e) Find the co-ordinates of the point D if C is the midpoint of BD (2)
- (f) Find the length of the interval joining BC (Leave answer in exact form) (2)
- (g)
 - i) Find the perpendicular distance of A from BC (Leave answer in exact form) (2)
 - ii) Hence, find the area of $\triangle ABC$ (2)
 - iii) Without calculating the area of $\triangle ABD$, find the ratio of the areas of the triangles ABC and ABD (1)
- (h) Find the co-ordinates of a point E such that $ABDE$ forms a parallelogram (1)
- (i) Find the area of the parallelogram $ABDE$. (2)

PART C**Start a new page.****16 Marks**

- (a) If α and β are the roots of the equation $2x^2 - 3x - 4 = 0$, find the value of
- (i) $\alpha + \beta$ (1)
- (ii) $\alpha\beta$ (1)
- (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ (2)
- (iv) $\alpha^2 + \beta^2$ (2)
- (b) Solve the equation $4^x - 9(2^x) + 8 = 0$ (2)
- (c) If $3x^2 - 5x + 6 = A(x - 2)^2 + B(x - 2) + C$ for all values of x , find values for A , B and C . (3)
- (d) Find the values of m for which the equation $x^2 + (m - 2)x + 4 = 0$ has equal roots; (2)
- (e) The quadratic expression $Q(x)$ is given by:
- $$Q(x) = (1 + m)x^2 + 4x + m - 1.$$
- Find the range of values of m for which $Q(x) > 0$ for all x . (3)

End of Exam ☺



Name: _____

Teacher: _____

Multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely, using a black pen.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct
↓

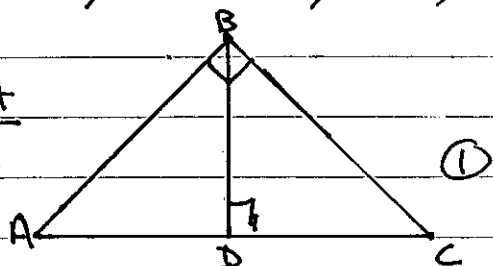
- Start here →
1. A ☐ B ☐ C ☐ D ☐
 2. A ☐ B ☐ C ☐ D ☐
 3. A ☐ B ☐ C ☐ D ☐
 4. A ☐ B ☐ C ☐ D ☐
 5. A ☐ B ☐ C ☐ D ☐
 6. A ☐ B ☐ C ☐ D ☐
 7. A ☐ B ☐ C ☐ D ☐

Multiple Choice

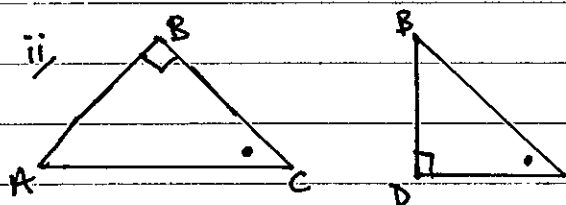
- 1, B D 2, A 3, A
4, D 5, D 6, C 7, D

PART A

a) i)

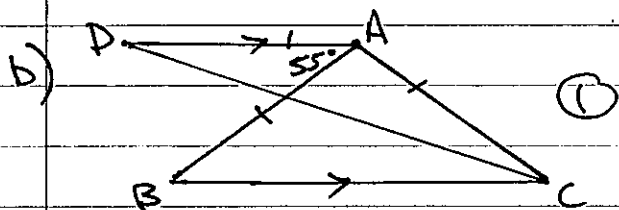


ii)



In $\triangle ABC$ and $\triangle BDC$
angle C is common (1)
angle ABC = angle BDC = 90° (given) (1)
 $\therefore \triangle ABC \equiv \triangle BDC$
(equiangular) (1)

iii $\frac{BC}{AC} = \frac{BD}{BC}$ (corresponding sides in ratio) (1)
 $BC^2 = AC \cdot DC$



ii) $\angle ABC = 55^\circ$ (alternate angle $AD \parallel BC$) (1)
 $\angle ABB = 55^\circ$ (base angles equal isos triangle) (1)

iii/ $\angle BAC = 180 - 2 \times 55$ (1)
 $= 70$ (angle sum of triangle)

$\therefore \angle ADC = \frac{180 - (70 + 55)}{2}$ (base angle of isos triangle ADC) (1)
 $= 27.5$

iv/ $\angle BCD = 55 - 27.5$ (1)
 $= 27.5^\circ$ (adjacent angle)

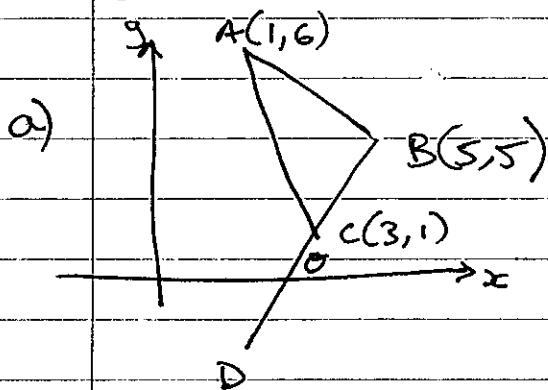
c) $\triangle AOB$ and $\triangle COD$
 $\angle AOB = \angle COD$ (vertically opposite) (1)
 $\angle BAO = \angle DCO = 90^\circ$ (given) (1)
 $BO = OD$ (given) (1)

$\therefore \triangle AOB \equiv \triangle COD$ (AAS) (1)

d) internal angle size = $\frac{(n-2) \times 180}{n}$
 $= \frac{(6-2) \times 180}{6}$ (1)
 $= 120^\circ$

$\therefore$ External angle $180 - 120^\circ$
 $= 60^\circ$ (1)

Part B.



b) i) $m = \frac{5-1}{5-3} = \frac{4}{2} = 2$ ①

c) $\tan \theta = 2$
 $\theta = \tan^{-1} 2$
 $\theta = 63^\circ$ ①

d) $y - 5 = 2(x - 5)$ ①
 $y - 5 = 2x - 10$ ①
 Eqn BD: $2x - y - 5 = 0$

e) across 2 up 4 $\therefore$ back 2 down 4
 ① various methods.

$D(3-2, 1-4)$

$D(1, -3)$ ①

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

f) $d = \sqrt{2^2 + 4^2}$ ①
 $BC = \sqrt{20}$

$BC = 2\sqrt{5}$ ①

3

$2x - y - 5 = 0$

g) i) $d = \frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$
 $= \frac{|2 \cdot 1 - 6 \cdot 1 - 5|}{\sqrt{2^2 + (-1)^2}}$ ①

$= \frac{|2 - 6 - 5|}{\sqrt{5}}$
 $= \frac{9}{\sqrt{5}} \text{ or } \frac{9\sqrt{5}}{5}$ ①

ii) $\frac{1}{2} \times 2\sqrt{5} \times \frac{9}{\sqrt{5}}$ ①

$= 9 \text{ units}^2$ ①

iii) As $\triangle ABC$ is half $\triangle ABD$
 ratio is 1:2 ①

j) from A to B across 4 down 1
 from D to E back 4 up 1
 $\therefore E(-3, -2)$ ①

k) Area of Parallelogram = 2 $\times$ Area $\triangle ABD$ ①

$\therefore A = 2 \times 18 \text{ units}^2$
 $= 36 \text{ units}^2$ ①

Part C

(a) for $2x^2 - 3x - 4$

i/ $\alpha + \beta = \frac{-b}{a} = \frac{3}{2}$ ①

ii/ $\alpha \beta = \frac{c}{a} = -2$ ①

iii $\frac{1}{\alpha} + \frac{1}{\beta}$

$$= \frac{\alpha + \beta}{\alpha \beta}$$
 ①

$$= \frac{\frac{3}{2}}{-2}$$

$$= \frac{3}{-4}$$
 ①

iv $\alpha^2 + \beta^2$

$$= (\alpha + \beta)^2 - 2\alpha\beta$$
 ①

$$= \left(\frac{3}{2}\right)^2 + 4$$

$$= \frac{9}{4} + 4$$

$$= 6\frac{1}{4}$$
 ①

b) $4^x - 9(2^x) + 8 = 0$

let $m = 2^x$

$$\therefore 2^{2x} - 9(2^x) + 8 = 0$$

$$m^2 - 9m + 8 = 0$$
 ①

$$(m-8)(m-1) = 0$$

$$m=8 \quad m=1$$

$$\therefore 2^x = 8 \quad 2^x = 1$$
 ①

$$x=3 \quad x=0$$

3/

c) $3x^2 - 5x + 6 = A(x-2)^2 + B(x-2) + C$

$$Ax^2 - 4Ax + 4A + Bx - 2B + C$$

$$Ax^2 - (4A-B)x + 4A-2B+C$$

equate co-efficients ①

$$A=3$$

$$4A-B=5$$

$$12-B=5$$

$$B=7$$

$$4A-2B+C=6$$

$$12-14+C=6$$

$$C=8$$
 ①

$$A=3 \quad B=7 \quad C=8$$

d) $x^2 + (m-2)x + 4 = 0$

$$b^2 - 4ac = 0$$
 ①

$$(m-2)^2 - 4 \cdot 1 \cdot 4 = 0$$

$$m^2 - 4m + 4 - 16 = 0$$

$$m^2 - 4m - 12 = 0$$

$$(m-6)(m+2) = 0$$

$$m=6 \text{ or } 2$$
 ①

e) $(1+m)x^2 + 4x + m-1 > 0$

$$\therefore b^2 - 4ac > 0$$
 ①

$$4^2 - 4 \cdot (1+m)(m-1)$$

$$16 - 4m^2 + 4$$

$$-4m^2 + 20 > 0$$

$$-4(m^2 - 5)$$

$$\therefore (m+\sqrt{5})(m-\sqrt{5}) < 0$$

$$m > \sqrt{5} \quad m < -\sqrt{5}$$
 ①